

RESEARCH INTERESTS

Human-AI Interaction, Social Computing, Conversational Agents, Communication

EDUCATION

University of Washington Ph.D., Human Centered Design & Engineering Advised by Prof. Gary Hsieh & Prof. Sean A. Munson	Seattle, WA 2023 - Present
Ewha Womans University B.S., Computer Science & Engineering, Minor in Statistics	Seoul, Korea 2019 - 2023
University of Toronto Undergraduate Exchange Program	Toronto, Canada 2021 - 2021

RESEARCH EXPERIENCES

Prosocial Computing Group, University of Washington Graduate Research Assistant (PI: Prof. Gary Hsieh) • Chatbot & Communication skills: Working on automated journalism and credibility frameworks to support scientific communication. • AI Journalism & Credibility: Working on automated journalism and credibility frameworks to support scientific communication. • Chatbot Personas & Credibility: Examining the effects of conversational agent design on search results.	May 2023 - Present
Intentional Traces Collective Lab, University of Washington Graduate Research Assistant (PI: Prof. Sean A. Munson) • Client–Clinician Match: Explored opportunities to support clinical communication and decision-making. • AI & Bibliotherapy: Investigated identity-matching interventions, focusing on the role of personal narratives and storytelling.	Jan 2024 - Present
Dynamic Graphics Project Lab, University of Toronto Undergraduate Research Assistant (PI: Prof. Khai N. Truong) • Creativity Support & Music Accessibility: Investigated opportunities for accessible music experiences for d/Deaf individuals using song signing.	Sep 2021 - Sep 2022
Human-Computer Interaction Lab, Ewha Womans University Undergraduate Research Assistant (PI: Prof. Uran Oh) • Voice Personas: Developed a search engine for labeled voice persona directories and a voice customization system. • Language Models & Patient Data: Applied bidirectional language models to classify electronic patient text data.	Jul 2020 - Sep 2023

PUBLICATIONS

Conference Papers

- [c.3] **Hyeonjeong Byeon**, Uran Oh, and Gary Hsieh. 2026. Understanding the Effects of Conversational Agent Personality on the Credibility of LLM-Based Conversational Search. *(To appear) In Proceedings of the 2026 ACM SIGIR Conference on Human Information Interaction and Retrieval (CHIIR '26)*. [paper]
- [c.2] Suhyeon Yoo, Georgianna Lin, **Hyeonjeong Byeon**, Amy Hwang, and Khai Truong. 2023. Understanding tensions in music accessibility through song signing for and with d/Deaf and Non-d/Deaf persons. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '23)*. [paper]

- [c.1] **Hyeonjeong Byeon***, Chaerin Lee*, Jeemin Lee, and Uran Oh. 2022. “A Voice that Suits the Situation”: Understanding the Needs and Challenges for Supporting End-User Voice Customization. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '22)*. [paper]

Posters and Extended Abstracts

- [p.4] **Hyeonjeong Byeon**, Seungjin Ha, and Uran Oh. 2023. AVOCUS: A Voice Customization System for Online Personas. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. [paper]
- [p.3] **Hyeonjeong Byeon**. 2021. Voice Creator: Giving Customized Voice to the Voiceless for Online Communication. In *Proceedings of the 23rd International ACM SIGACCESS Conference on Computers and Accessibility Student Research Competition (ASSETS' SRC 21)*. [paper]
- [p.2] **Hyeonjeong Byeon**, and Uran Oh. 2021. Comparing Performance of Pre-trained Bidirectional Language Models. In *Proceedings of HCI Korea Conference 2021*. [paper]
- [p.1] **Hyeonjeong Byeon**, Soohyun Yeo, and Uran Oh. 2021. Voice Creator: A Vocal Customization Web Application Prototype. In *2021 Korea Information Processing Society Spring Conference No. 28, Vol. 1*. [paper]

GRANTS & AWARDS

Doctoral Student Research Grant	Dec 2025
University of Washington, Received grants to support projects for LLM-infused System Intervention Design	
Social Entrepreneurship Fellow	April 2025
Population Health Initiative, Leading the Virtual Study Assistant for Potential Research Participants Project	
Principal Candidate, Fulbright Graduate Student Program Award	Sep 2023
Korean-American Educational Commission, awarded up to \$80,000 for graduate studies (<i>declined</i>)	
Finalist, Doctoral Study Abroad Scholarship	Aug 2023
Korean Foundation for Advanced Studies (KFAS), selected as a finalist	
ITRC Student Creative Research Project	March 2022 - Dec 2022
South Korean Ministry of Science and ICT, awarded \$7,600 of research grant	
Capstone Poster Session Presentation	Dec 2022
Ewha Womans University, 2nd place in Computer Science department, co-awarded \$570	
SW Entrepreneurship Competition	Nov 2022
Ewha Womans University, 4th place, co-awarded \$75	
Google Student Travel Grants	May 2022
Google, awarded \$2,820 for attending ACM CHI 2022	
Ewha Future Design Scholarship	Dec 2022, April 2022
Ewha Womans University, awarded \$3,240 for excellence in career path planning	
3rd place (Bronze Medal), Student Research Competition	Oct 2021
ACM SIGACCESS Conference on Computers and Accessibility (ASSETS' 21) Student Research Competition, awarded \$200	
Presidential Science Scholarship	June 2021
Korea Student Aid Foundation (KOSAF), awarded cost of attendance for 2 years Additional \$2,200 per semester for academic excellence	
Best Paper Award	Jan 2021
Proceedings of HCI Korea Conference 2021, Seoul, South Korea	
Undergraduate Research Opportunity Program Research Grant	April 2022, June 2020
Ewha Womans University, awarded \$250, \$450 respectively	

HOKMA Career Path Exploration Scholarship Grant

Oct 2019

Ewha Womans University, co-awarded \$1,100 for excellence in CS career path exploration activities

PATENT

Systems and Methods for Customizing Voices for Personas in Virtual Spaces

Dec 2022

Uran Oh, Hyeonjeong Byeon, Seungjin Ha, 10-2022-0179981. *Patent pending.*

AVOTAR: A Voice Customization System for Online Personas

Nov 2022

Uran Oh, Hyeonjeong Byeon, Seungjin Ha, South Korea, P2022-0160KR. *Patent pending.*

TEACHING EXPERIENCE

Visual Communication (HCDE 508)

Designing For Behavior Change (HCDE 538)

Usability Studies (HCDE 517)

User Experience Prototyping Techniques (HCDE 451)

Introduction to User Research (HCDE 313)

SERVICE

Review Service

2024 -

ACM CHI (2026 -), ACM CHI Late-Breaking Work (2024 -), IASDR (2025)

Student Admission Reviewer

Dec 2023

Participated as a student reviewer for UW HCDE Ph.D. admission

Student Volunteer, The Blue Tree Foundation (NGO)

Feb 2022 - Feb 2022

Researched and translated articles about school violence and cyberbullying for campaigns

Member, EDOC (Ewha DO Coding)

Jan 2020 - Jun 2020

Prepared for ACM-ICPC and participated in group study sessions focusing on data structures and algorithms

Student Volunteer, Seoul Metropolitan Office of Education

Sep 2019 - Nov 2019

Taught programming to North Korean defector students as part of the “Unplugged Coding” program

Member, Like Lion

Mar 2019 - Dec 2019

Completed a year-long full-stack development course.

Back-end development of the university festival web application [Web Demo] (*closed*)

RELEVANT SELECTIVE COURSEWORK

Natural Language Processing (CSE 517)

Reproduced an EMNLP 2024 paper and conducted additional generalizability experiments [code]

Core Concepts in Decision Making (PSYCH 521)

Dual-process theories of reasoning, heuristics and biases, nudging, and decision-making with AI support

Applied Social Statistics (SOC 504)

Regression modeling, hypothesis testing, and applied statistical inference

Communication in Small Groups (COM 577)

Seminars on group interaction, social influence, norms, trust, identity, and performance

SKILLS

Research: Empirical studies, Statistical modeling, Inferential Statistics, Survey, Prototyping

Programming: Python, C/C++, Java, HTML/CSS, JavaScript

Data analysis: R, SAS, SQL